MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – TEST (SET – B)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM: 25**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. This question paper contains 11 compulsory questions divided into four Sections: A, B, C, and D.
2. **Section A** comprises 5 Multiple Choice Questions (MCQs) of 1 mark each (Q1 to Q5).
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each (Q6 and Q7).
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each (Q8 to Q10).
5. **Section D** comprises 1 Long Answer (LA) question of 5 marks (Q11). An internal choice is provided.
6. Draw neat, labeled geometric figures wherever applicable. Use of calculators is strictly prohibited.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. The ratio of the length of a pole and its shadow on the ground is $1 : \sqrt{3}$. The angle of elevation of the Sun is: [1]

(A) $90°$ (B) $60°$ (C) $45°$ (D) $30°$

Q2. A vertical pole 10 m long casts a shadow of length 5 m on the ground. At the same time, a tower casts a shadow of length 12.5 m on the ground. The height of the tower is: [1]

(A) 25 m (B) 20 m (C) 15 m (D) 10 m

Q3. A kite is flying at a height of 150 m from the ground. It is attached to a string inclined at an angle of $30°$ to the horizontal. The length of the string is: [1]

(A) 150 m (B) $150\sqrt{3}$ m (C) 300 m (D) $300\sqrt{3}$ m

Q4. An observer 1.8 m tall stands away from a chimney at a distance of 38.2 m along the ground. The angle of elevation of the top of the chimney from his eyes is $45°$. The height of the chimney is: [1]

(A) 40 m (B) 38.2 m (C) 36.4 m (D) 41.8 m

Q5. A wire is attached from a point A on the ground to the top of a pole BC, making an angle of elevation of $60°$. If $AB = 5\sqrt{3}$ m, then the length of the wire is: [1]

(A) 10 m (B) $10\sqrt{3}$ m (C) 15 m (D) $5\sqrt{3}$ m

SECTION B: Very Short Answer Questions (2 × 2 = 4 Marks)

Q6. The angle of elevation of the top of a tower from a point on the ground which is 30 m away from the foot of the tower, is $30°$. Find the height of the tower. [2]

Q7. A 1.5 m tall boy is walking away from the base of a lamp post which is 12 m high, at the speed of 2.5 m/sec. Find the length of his shadow after 4 seconds. [2]

SECTION C: Short Answer Questions (3 × 3 = 9 Marks)

Q8. A man standing on the deck of a ship, which is 10 m above water level, observes that the angle of elevation of the top of a hill is 60° and the angle of depression of the base of the hill is 30°. Find the distance of the hill from the ship and the height of the hill. [3]

Q9. As observed from the top of a lighthouse 60 m high from sea level, the angles of depression of two ships are 45° and 60°. If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships. (Use $\sqrt{3} = 1.732$) [3]

Q10. An aeroplane when flying at a height of 3125 m from the ground passes vertically below another plane at an instant when the angles of elevation of the two planes from the same point on the ground are 30° and 60° respectively. Find the distance between the two planes at that instant. [3]

SECTION D: Long Answer Question (1 × 5 = 5 Marks)

Q11. The angle of elevation of an aircraft from a point A on the ground is 60°. After a flight of 30 seconds, the angle of elevation changes to 30°. The aircraft is flying at a constant height of $3600\sqrt{3}$ m. Find the speed of the aircraft in km/h. [5]

— OR —

A person standing on the bank of a river observes that the angle of elevation of the top of a tower on the opposite bank is 60°. When he moves 30 m away from the bank, he finds the angle of elevation to be 30°. Find the height of the tower and the width of the river. (Take $\sqrt{3} = 1.732$) [5]

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